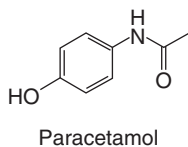


8.4 Measurement Uncertainty: Special Issues

8.4.1 Influence of the Calibration Model



The use of a calibration model to predict concentrations seems perfectly normal and expected for analysts. However, the question remains whether the chosen calibration model can be a significant source of uncertainty. A publication on the validation of a paracetamol method for drug control may help to answer this question [14]. The following table summarizes the characteristics of the PARACETAMOL dataset.

In the original publication [14], it is not stipulated if a replicate consists of applying the entire analytical operating procedure on unique test portions or whether replicates are limited to repeated measures on a single, fully prepared test portion. We will consider a replicate a “full replicate” as defined in Section 8.4.3. Complete dataset description is reported in Table 8.3.

Using the calibration experimental data, it is possible to fit two types of models. The question of selecting the best calibration model has already been explained in Section 8.1 and resolved using the MAP procedure. When X is the paracetamol content of the calibration solutions expressed in mg/ml, and Y is the instrumental response expressed in absorbance units, two models can be fitted:

Linear curve

$$Y = a_0 + a_1X + e$$

Table 8.3 PARACETAMOL – description of the dataset.

Title	PARACETAMOL
Reference	[14]
Measurer	Paracetamol concentration in a drug, expressed in mg/ml.
Method	UHPLC (ultrahigh-performance liquid chromatography) coupled with UV detection
Validation area	Between 25 and 150% of the nominal value of the finished product of 3.25 mg/ml (0.8125 to 4.875 mg/ml)
Acceptance interval	±5%. Value often used for medicine quality control
Validation materials	4 synthetic validation materials containing 0.8125, 1.95, 3.25 and 4.875 mg/ml respectively prepared by weighting.
Validation design	Series ($I = 3$), replicates/series ($J = 5$), levels ($K = 4$)
Calibration design	Series ($I' = 3$), replicates/series ($J' = 3$), levels ($K' = 6$)
Total number of measures	54 measurements on calibration solutions 60 measurements on materials with known contents.

Table 8.4 PARACETAMOL – calibration model coefficients.

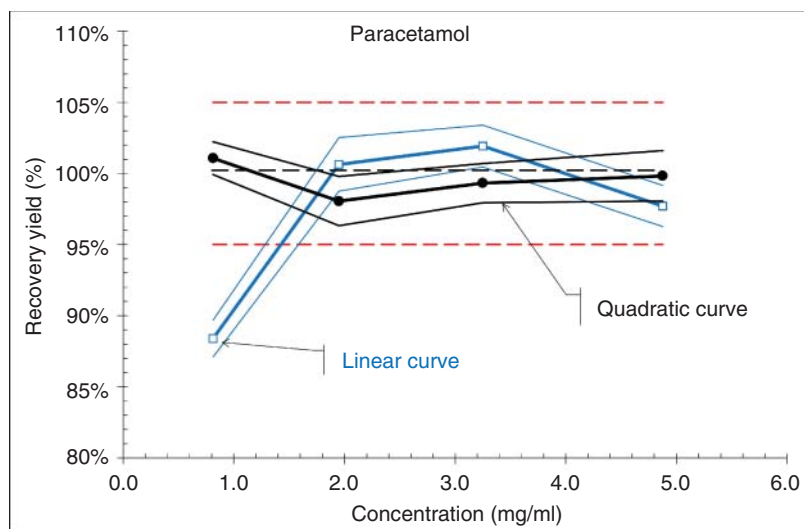
Series	Model	a_0	a_1	a_2	r^2
Series 1	Linear	16.984	52.843		0.996
Series 2		16.562	53.075		0.996
Series 3		16.205	53.319		0.995
Series 1	Quadratic	0.712	67.862	−2.641	0.997
Series 2		1.550	66.932	−2.436	0.999
Series 3		0.826	67.514	−2.496	0.988

Quadratic curve

$$Y = a_0 + a_1X + a_2X^2 + e$$

Table 8.4 combines the coefficients for each serial (or daily) model. In this example, coefficients are stable from one day to another. The opposite conclusion was made with the THEOPHYLLINE study. However, selecting the best calibration model is difficult just by looking at the coefficients of determination and is even impossible in this case.

The MAPs obtained using the two models are plotted in Figure 8.9. Once more, the decision is easy as it is obvious that the linear model is unsatisfactory while the quadratic calibration model should be applied. The conclusion is evident, the method accuracy profile can be successfully used to achieve the final optimization of the method.

**Figure 8.9** PARACETAMOL – method accuracy profiles using two calibration models on the same data.

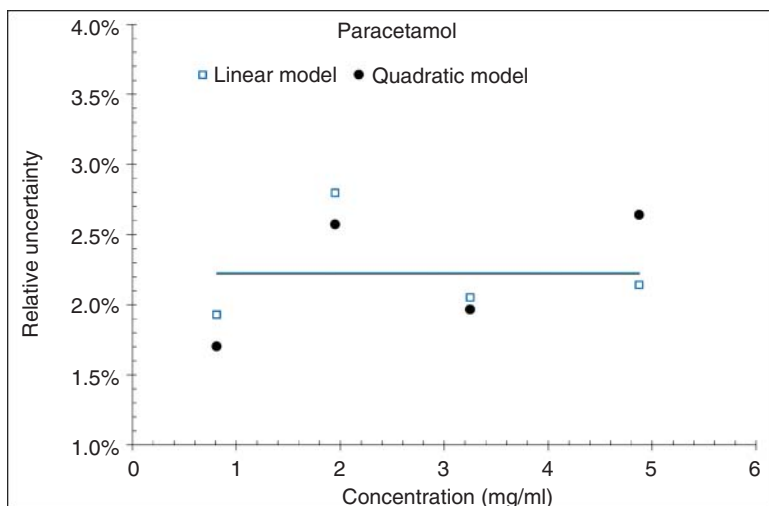


Figure 8.10 PARACETAMOL – uncertainty functions for the two calibration models. White squares: linear, gray circles: quadratic model.

Considering MU, it is also possible to compute the corresponding values for each model and concentration with the statistical parameters already used to build the two profiles. These relative uncertainty values are illustrated in Figure 8.10 as a function of the paracetamol concentration. Since the validation range is narrow, relative uncertainty does not significantly vary. The most appropriate way to fit the relative uncertainty function is the straight line parallel to the concentration axis. Whatever the calibration model, linear or quadratic, both uncertainty functions overlap. This means that the relative MU is constant over the whole validated range and equal to 2.2% on average.

Finally, for the PARACETAMOL study, the calibration model has no influence on the MU. Several reasons may explain this fact: the quite high concentrations of the validation materials ranging from 1.0 to 5.0 mg/ml; by definition, MU only considers the dispersion of results and not the bias that is assumed corrected, while a flawed calibration model mainly impacts the bias.

This is not a general conclusion, and there are other situations where an inappropriate calibration model may alter the MU. The same comparison can be made with the THEOPHYLLINE dataset, where several uncertainty functions can be built with different calibration models.

Figure 8.11 compares the uncertainty functions derived from inverse-predicted results when the quadratic model is calculated with the OLS or WLS regression method. It seems logical to assume that the best model reduces the MU. This assumption can be verified here as the WLS model produces smaller MU, mainly at low concentrations. Several reasons can also be mentioned to explain this difference: the low concentrations of the validation materials ranging from 0.05 to 10.0 $\mu\text{g/l}$ (i.e. 10^6 times lower than paracetamol); the vicinity to LOQ; the daily variability of the calibration model coefficients, and so on.

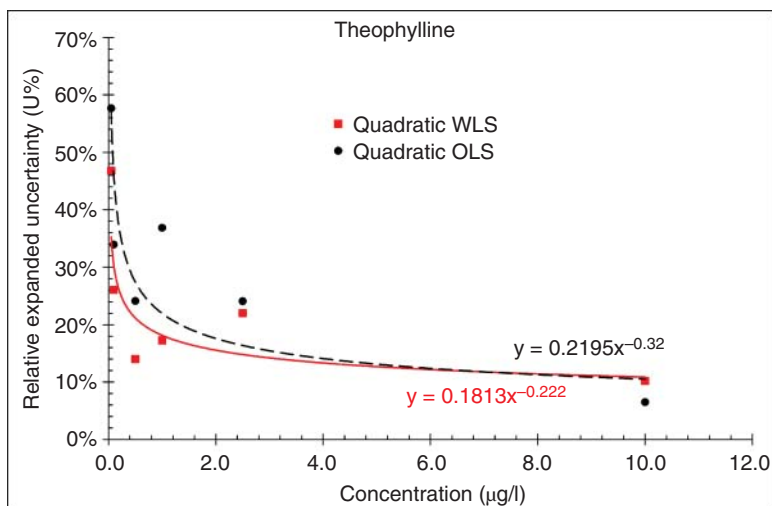


Figure 8.11 THEOPHYLLINE – comparison of uncertainty functions obtained with two calibration models.

8.4.2 Uncertainty of Corrected Results

To fully apply the GUM general procedure, before providing an estimate of MU, it is stated in clause 3.2.4:

“... it is assumed that the result of a measurement has been corrected for all systematic effects recognized as significant and that every effort has been made to identify them...”

This clause implies that, in most cases, a correction is possible by modifying the calibration function. However, if this is true for physical measurements, it is not often the case in the analytical sciences. The method accuracy profile (MAP) can be a tool for identifying the eventual bias of the method, and it is possible to graphically check if the trueness of the method is satisfactory.

When the bias is considered a significant source of inaccuracy, it is necessary to establish a correction factor applicable to the inverse-predicted concentrations. In this case, the uncertainty function should be modified accordingly. When considering the theophylline MAP illustrated by Figure 5.5 it is possible to identify a significant bias at low concentration. Table 5.4 reports the values of the relative bias as a function of the concentration.

Let us remember that the relative bias is equal to 100% – recovery yield. For instance, for a concentration of 0.1 µg/l the recovery yield is 88% and the relative bias 12%. The latter diminishes very quickly and becomes negligible in regard to the MU in the rest of the validation range.

THEOPHYLLINE: Relative bias as a function of concentration

Levels (µg/l)	0.05	0.1	0.5	1	2.5	10
Relative bias (%)	17	12	4	0	1	4

For this example, because it is estimated that the LOQ is about 0.19 µg/l, it seems unnecessary to correct the data for the relative bias. But, once the need to apply a correction factor has been identified and validated, it becomes a part of the operating procedure and must be applied systematically to all new measurements. This statement must be qualified by two remarks:

- The correction factor is specific to a given matrix.
- The correction factor may vary with the concentration level.

The correction factor is a new source of uncertainty that must be included in the estimate of the MU for the corrected measurement value. It should be remembered that correcting biased measurements is a perfectly acceptable practice within ISO 17025 accreditation. It must be scientifically justified and reported to the end-user.

Reviews of different methods applicable to the estimation of the MU of a corrected measurement are available in [15, 16]. The solution presented below is developed in agreement with the GUM recommendation described in Appendix F.2.4.5 of GUM (2018 version). It is stated that “when corrections are not applied from a calibration curve...” an approach described as “relatively simple for this problem, consistent with GUM principles” can be applied. The GUM procedure consists of calculating an average correction factor for a given measurand. Let us consider the data collected for building a MAP.

The measurand is the assigned target concentration X_k of validation material. Following the proposed notation, Z_{ijk} is the inverse-predicted concentration of replicate j ($1 \leq j \leq J$) in series i ($1 \leq i \leq I$) for the validation material k ($1 \leq k \leq K$). The absolute individual bias δ_{ijk} is defined by Equation (8.9). The average correction factor for a given value of X_k is defined by the arithmetic mean of all biases $\overline{CF_k}$. The corrected predicted concentration Z_{ijk}^* is defined by Equation (8.11). It is essential to utilize an additive correction factor, whereas a multiplicative one would increase or decrease the measurement value and artificially modify the MU. The standard variance of the average correction factor is defined by Equation (8.12).

The bias of an inverse-predicted concentration

$$\delta_{ijk} = Z_{ijk} - X_k \quad (8.9)$$

Average correction factor for level k

$$\overline{CF_k} = \frac{\sum_i \sum_j \delta_{ijk}}{IJ} \quad (8.10)$$

Corrected measurement value

$$Z_{ijk}^* = Z_{ijk} - \overline{CF_k} \quad (8.11)$$

The standard variance of the average correction factor

$$u^2(\overline{CF_k}) = \frac{\sum_i \sum_j (\delta_{ij} - \overline{CF_k})^2}{(IJ - 1)} \quad (8.12)$$

Applying the law of propagation of uncertainty for a sum of input quantities, the combined standard uncertainty of a corrected measurement is: